

# A Random Talk on Random Walks

Garrett Kepler

April 7th, 2026



# Outline

① Probability Theory

② Gerrymandering

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We'll cover some basics today. But, I am always available for discussion!

- Probability – Weak Law of Large Numbers and Markov Chains
- Gerrymandering – How we can actually generate random redistricted maps
- Mathematical History – My own and others

① Probability Theory

② Gerrymandering





# Questions

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- What is the *average* of  $\{1, 2, 3, 4, 5, 6\}$ ?
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## Complicatedly

- The average is the sum  $\frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$
- The *average of the values we expect to see* is 3.5  
( $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$  where  $X$  is the outcome of a dice roll)

# Let's Roll Some Dice!

What we will do:

- Raise your hands (or shout out) with a number 1 through 6
- I will plot the averages

<https://garrett-kepler.github.io/resources/LLN/>

# Guide 1

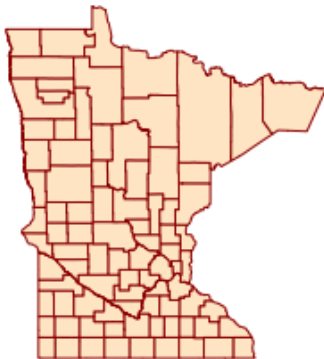


Figure: Gary Isaacs (8th-9th grade)

# Weak Law of Large Numbers (WLLN)

## Intuitively

The average should be close to the expected value eventually. That is, if we roll a dice a lot, we should see something close to 3.5 as we continue to roll.



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## Complicatedly

If  $X_1, X_2, \dots$  are a sequence of independent identically distributed random variables with  $\mathbb{E}[X] < \infty$ , then for any  $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[ \left| \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_i] \right| < \epsilon \right] = 1$$



## Guide 2



Figure: Jaime Garcia (12th grade)

# WLLN with Dependence

What if instead of asking you to give me a random number, I asked you to raise a random number that your neighbor *didn't* raise?



## WLLN with Dependence

What if instead of asking you to give me a random number, I asked you to raise a random number that your neighbor *didn't* raise?

This time, whatever the person before you says (a random number 1 through 6), give me a random number between 1 through 6 they *didn't* say.

<https://garrett-kepler.github.io/resources/LLN/>

## A New School of Thought



**Figure:** (Left) Pavel Nekrasov: firm believer WLLN only applies to independent events. (Right) Andrey Markov: prover that WLLN applies to dependent events too!

# WLLN with Dependence

## Intuitively

The beautiful result Markov proved is that we actually don't need independent events! With a little assumptions we can confidently say the average will get close to the expected value.

# WLLN with Dependence

## Intuitively

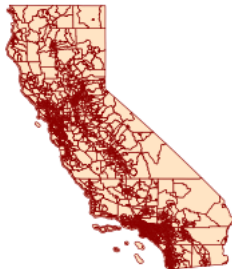
The beautiful result Markov proved is that we actually don't need independent events! With a little assumptions we can confidently say the average will get close to the expected value.

## Complicatedly

If  $X_1, X_2, \dots$  is a positive Harris recurrent Markov chain with unique invariant distribution  $\pi$  and  $f$  is a real-valued function on the state space, then

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{a.s.} \mu = \mathbb{E}_\pi f(X) = \int f(x) \pi(dx)$$

## Guide 3



**Figure:** Dr. Andrea Arauza Rivera (Junior-Senior year undergrad)

## Guide 3 (continued)

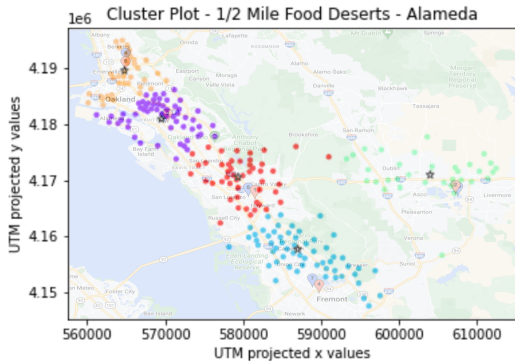


Figure: Some work with Dr. Arauza Rivera (including graduation)

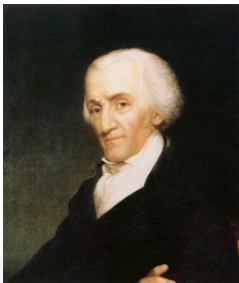


① Probability Theory

② **Gerrymandering**



# History of Gerrymandering



**Figure:** Eldridge Gerry and his “salamander district” in 1812 Massachusetts: ([Click here for full map](#))

A portrait of John Jay, an older man with white hair, wearing a dark coat and a white cravat. He is looking slightly to the right of the viewer.

## Theorem

Proof.

5

# The Maps We See

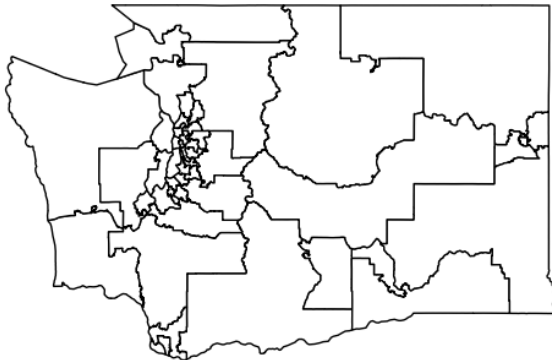
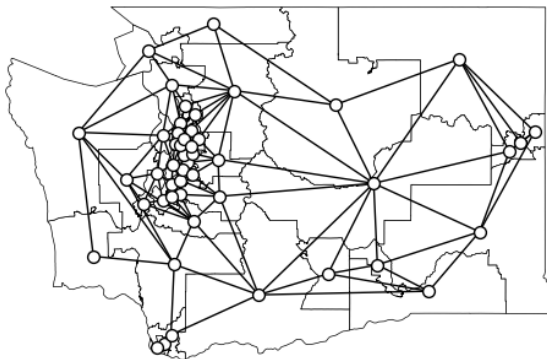


Figure: Washington's 2020 Legislative Districts

# The Maps We Deal With

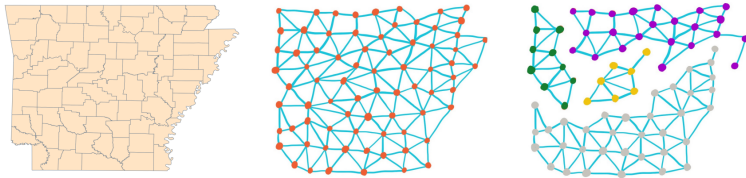
In practice, it is very difficult with maps themselves. So, we simplify them.



**Figure:** Washington's 2020 Legislative Districts and corresponding dual graph

# The Maps We Deal With

Once we have the dual graphs, we can start thinking about district assignments.



**Figure:** Example of a dual graph of Arkansas with its district assignment as a partition

# Gerrymandering through Experience

Among the best ways to intuit how gerrymandering works is quite possibly by trying it for yourself.

We will be given voting information. What we will do:

- Try and form districts that advantage our party
- Try and form districts that disadvantage an opponent's party

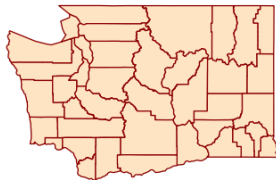
<https://garrett-kepler.github.io/resources/gerry/>

## Try It Yourself



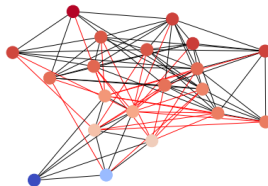
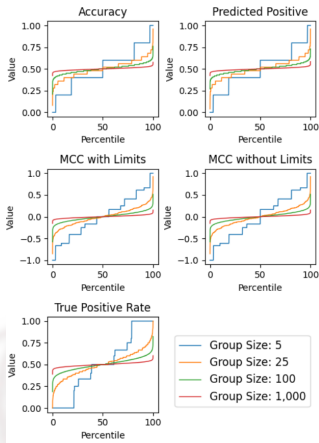


## Guide 4



**Figure:** Dr. Daryl DeFord (first-fourth year grad school)

## Guide 4 (continued)



**Figure:** Some work with Dr. DeFord: (left) machine learning metrics bias (right) similar spectral partitioning using MCMC

# Gerrymandering through “Definition”

*“Gerrymandering, defined in the contexts of representative electoral systems, is the political manipulation of electoral district boundaries to advantage a party, group, or socioeconomic class within the constituency.”* - Wikipedia circa 2:53 PM April 6th, 2026 (For Link Click Here)

## Question

Okay, but how do we know when someone is trying to advantage a party/group?

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Okay, but how do we know when someone is trying to advantage a party/group?

## Answer (\*)

We could draw all the possible maps and compare to the map someone wants to enact!

# Womp Womp :(

## Answer (\*)

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Actually, this highly impractical...

```
In [19]: start = time.time()
allperms = list(permutations(partition))
print('Runtime: '+str(time.time()-start))

-----
MemoryError                                Traceback (most recent call last)
<ipython-input-19-04af173883b2> in <module>
      1 start = time.time()
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MemoryError:
```

**Figure:** Memory error for just trying to calculate all 4 partitions of grid from game with 16 nodes (circa 5:12 PM April 6th, 2026)

# Data Constraint

16 is comparatively a small amount! For example:

- Minnesota has 201 legislative districts
- Washington has 49 legislative districts
- California has 172 legislative districts
- They are not the only ones!

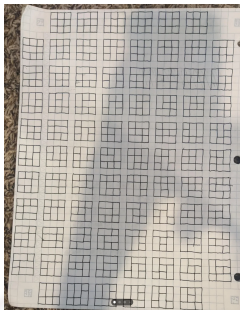
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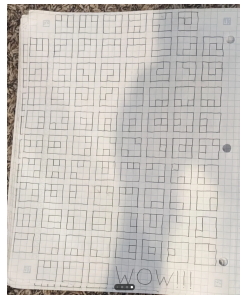
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Trying to generate every possible map would be an insane feat of computing that we do not have time for. Maps need to be decided on way sooner than the time it would take to compute.

# Perspective



... 15 pages later ...



**Figure:** For perspective, here are the connected partitions of the 3 by 3 grid enumerated by a Reddit user (17 photos total!)



# The Recombination Approach

What we do in practice is much more clever:

- Start a map with some districts
- Perturb the partitioning a little bit
- Calculate whatever you like and repeat

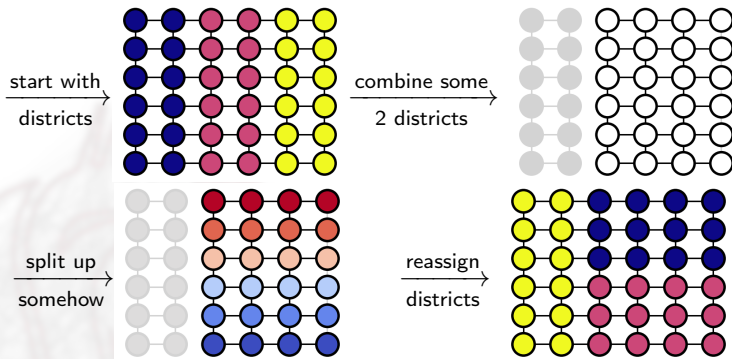
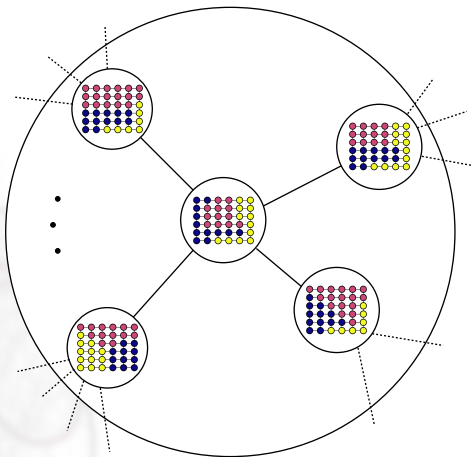


Figure: An example of what we're doing

# Metagraph

Now, what we get to do is move around the space of districted maps. Moreover, the way we do it guarantees Markov's LLN applies!



# LLN for Maps

## Intuitively

If we recombine and split districts cleverly, we can get a *good* set of maps that we can compare to. That way, we can detect gerrymandering when it occurs!



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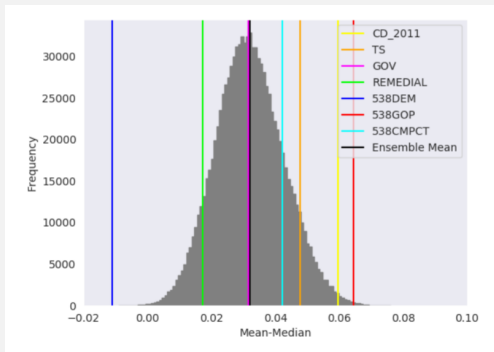
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## Complicatedly

Ideally, we will define an aperiodic Markov chain with one recurrent class such that the Ergodic theorem applies and we have a unique stationary distribution of districted maps. Then, we can get representative samples from the stationary distribution.

# Ensemble Analysis

This idea is what we call ensemble analysis: Draw a lot of *relevant* maps and compare them to proposed maps.



**Figure:** Comparison of scores from recombination-based maps (grey) with scores of politically proposed maps (vertical lines)

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- And more!

# “Fairness”

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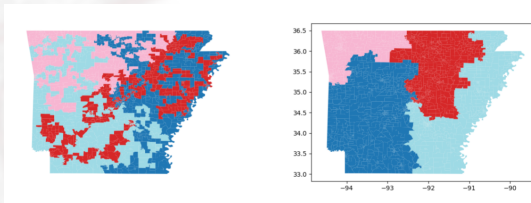
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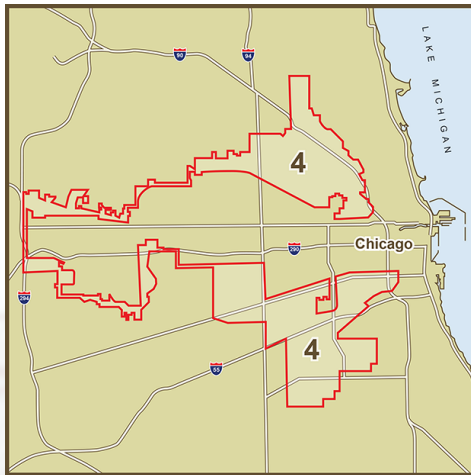
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While a map may not pass the “eyeball” test, it may be the best option.



# “Fairness”



**Figure:** For example, the famous “Horseshoe District” from Chicago. Making this district score well in compactness would make it score poorly in majority-minority ratio.

# Political Science vs. Mathematics

As a political scientist, you may ask questions like:

- What are the best ways to accurately represent a group in an election?
- How might we protect against gerrymandering through legislature?





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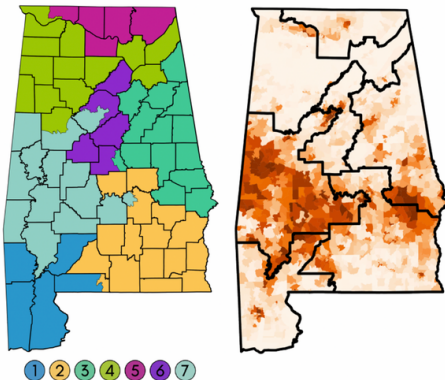
- What are the best ways to accurately represent a group in an election?
- How might we protect against gerrymandering through legislature?

As a mathematician, you are given constraints your random maps must satisfy. Now that you know them, try and:

- generate random maps that satisfy the constraints (as many as is practically possible)
- provide statistical statements about the space of possible graphs for court cases where gerrymandering is in question

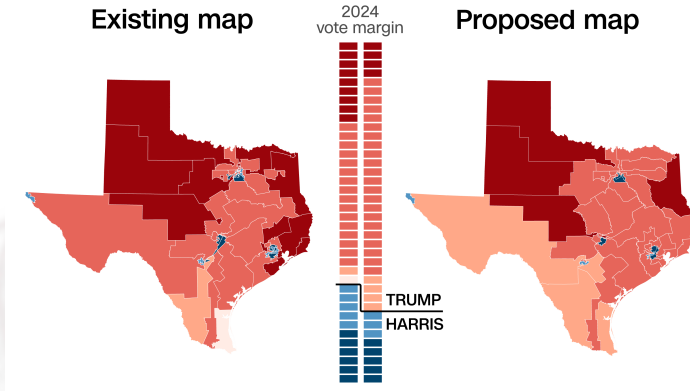
## These Questions Still Arise

In 2023, the Supreme Court ruled that Alabama had racially gerrymandered.



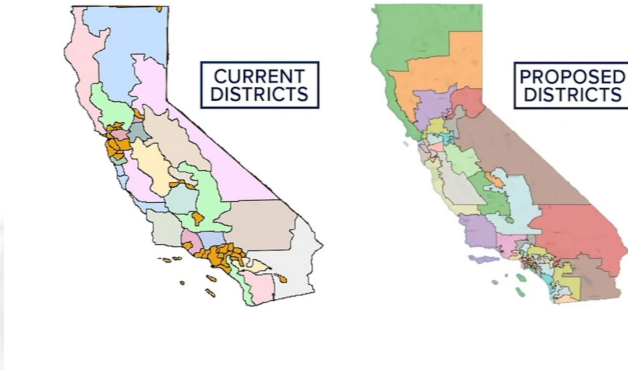
# These Questions Still Arise

In 2025, the Supreme Court ruled that Texas had not racially gerrymandered.



# These Questions Still Arise

In 2025, California *voted* that they redraw their districted map.

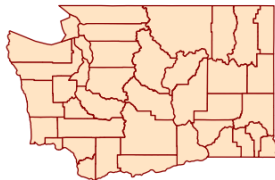


# These Questions Still Arise (and There May Be No Answer)

We are not be able to make the perfect map, but we can try and detect bad maps and do better.



## Guide 5



**Figure:** Dr. Matthew Hudelson (fourth-fifth year grad school)

## Guide 5 (continued)

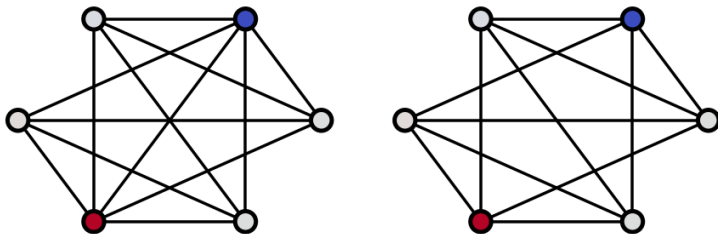


Figure: Some work with Dr. Hudelson: use of eigenvectors in graph reconstruction and perturbation

# Conclusion & Homework

## Key Takeaways:

- Generating random maps requires some finesse
- Generating “fair” random maps requires even more
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## Things to contemplate:

- Fairness in redistricting
- The people in your life that make it easier/better/interesting/peaceful/etc.

# Resources

For the curious, here are some redistricting-focused resources:

- “*Political Geometry*” – Textbook on mathematical redistricting methods  
<https://data-democracy.org/gerrybook.html>
- “*Redistricting Data Hub*” – Dataset resource with redistricting-oriented data <https://redistrictingdatahub.org/>